Brian Cahill

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Work Experience

Amazon Robotics North Reading, MA
L4 Solutions Design Engineer | Lead Solutions Deployment Engineer, Robotic Storage Technology April 2024 – Current

- Lead end-to-end deployment for all North American ToteASRS new building and retrofit projects, managing 6 concurrent fulfillment center deployments with accountability for survey packages, production maps, IP address tables, and deployment documentation
- Developed Python automation tools eliminating 20+ hours of manual work per designer per deployment: Wireless Access Point placement script (reduced 8+ hour task to <5 minutes), Excel macros for survey package formatting and IP address table generation (4-hour task to <5 minutes)
- Created an AI LISP agent that easily integrates with AutoCAD and automates manual tasks. 2 week beta test with 4 designers has decreased CAD time by 50% equating to ~500 hours per designer annually
- Delivered two critical site survey packages in 2 business days (typical timeline: 2 weeks), saving $192k in remobilization costs alone and preventing project delays and even more substantial financial losses
- Wrote a LISP program analyzing 750k+ sq. ft AutoCAD drawings saving thousands of collective hours annually in drawing analysis and on-site rework due to preventable errors made during design
- Built comprehensive ToteASRS Deployment Wiki establishing standardized processes, SLA frameworks, and quality control checklists saving 20+ hours per new designer per deployment
- Achieved 100% on-time delivery rate with 0 critical deployment errors across 50+ production-level map deployments and 4 complete floor survey packages
- Led Solutions Team Office Hours as team resource for the RST organization and led weekly Deployment Sync as Solutions subject matter expert, training L5 Senior IDEs, L4 IDEs, and CAD technicians on critical deployment processes
- Received 2 promotions in <2 years because of aptitude and work ethic

AcousticaBio Cambridge, MA
Automation Engineering Co-op January 2023 – August 2023

- Designed numerous components of an acoustophoretic printer capable of emitting a 250g force
- Created a python/ImageJ script to analyze videos and pictures of microparticles and produce statistics and distribution plots of printer performance
- Prepared and executed microparticle concentration and size experiments on an acoustophoretic printer

Raytheon Missiles & Defense Andover, MA
Operations Engineering Co-op July 2021 - January 2022

- Created an AutoCAD floor plan documentation standard for the Andover Industrial Engineering team
- Built a QR scanner log in VBA for floor operators to easily track Work-in-Progress and minimize lost material
- Led the AutoCAD effort to reorganize over 100,000 sq. ft of factory space on the Andover Campus communicating between multiple team leaders and identifying unrecognized areas of opportunity

Education

Northeastern University Boston, MA December 2023
Bachelor of Science in Mechanical Engineering, Minor in Data Science *GPA*: 3.8
Awards: Magna Cum Laude, Dean's List (all semesters), 1st place in Mechanical Engineering Capstone track
Senior Capstone Project Metal Powder Aerosolizer: Created a novel method to aerosolize fine powders using vibrating ERM motors and a pressure differential system for cold spray 3D printing applications

Skills & Certifications

Programming & Automation: Python, AutoLISP, SQL (MySQL), VBA, MATLAB, C++, IJM (ImageJ Macro)
CAD & Engineering Software: AutoCAD, SolidWorks (CSWA), Kiva Map Editor, ASM (Amazon Site Maps), FEA, Arduino
Data & Analysis: Excel (Advanced Macros), Statistical Modeling, ImageJ, Data Visualization
Certifications & Technical: DoD Secret Security Clearance, 3D Printing (FDM/SLA), Laser Cutting, Soldering

Interests

Enthusiastic about hockey, golf, football, and baseball; enjoy mathematical puzzles, personal fitness, and tv shows